

Mixed Methods and Sampling

120 minutes · 6 self-check questions

Jesse Anak · Institute of Computational Biology and Bioinformatics

How quantitative and qualitative strands are combined deliberately rather than decoratively, and the full range of sampling strategies with the arithmetic of sample size.

Learning objectives

- Describe convergent, explanatory sequential and exploratory sequential designs
- Explain what integration means and why it is the defining feature of mixed methods
- Distinguish probability from non-probability sampling and state when each is appropriate
- Calculate a sample size for a prevalence study and for comparing two proportions
- Explain power, effect size and the consequences of an underpowered study
- Adjust a sample size for non-response and design effect

Mixed methods designs

- Convergent parallel — both strands run at the same time and are compared at interpretation; used when each answers a different facet of one question
- Explanatory sequential — quantitative first, then qualitative to explain surprising or unexplained results
- Exploratory sequential — qualitative first to develop an instrument or identify constructs, then quantitative to test at scale
- Embedded — one strand sits inside a larger design, such as a process evaluation within a trial
- Integration is the defining feature: strands must be brought together, not merely reported in adjacent chapters
- Integration techniques include joint displays, following a thread, and merging in a triangulation matrix

IN PRACTICE

A study that runs a survey and some interviews and reports them separately is not mixed methods; it is two studies in one document. The examiner's question is always: what did you learn from combining them that neither strand gave you alone?

Probability sampling

- Simple random — every unit has an equal chance; requires a complete sampling frame
- Systematic — every k th unit from a random start; simple in the field but fails if the list has a periodic pattern
- Stratified — the population is divided into strata and sampled within each; improves precision and guarantees representation of small subgroups
- Cluster — whole groups are sampled; cheaper when the population is geographically dispersed, but less precise
- Multistage — combines the above, and is what national surveys such as the DHS actually use
- Only probability sampling supports statistical generalisation to the population

Non-probability sampling

- Convenience — whoever is available; cheapest and weakest, with unknown bias
- Purposive — chosen for a defined characteristic; the standard for qualitative work
- Quota — fills preset category targets non-randomly; resembles stratification without randomisation
- Snowball — participants recruit others; necessary for hidden populations, but stays within networks
- Consecutive — every eligible patient over a defined period; the strongest non-probability option in clinical settings
- These cannot support statistical generalisation, and analyses reporting confidence intervals from them should be read cautiously

Sample size

- For a single proportion: $n = Z^2 p(1-p)/d^2$, where Z is 1.96 for 95% confidence, p the expected proportion, d the desired margin of error
- Worked example: $p = 0.30$, $d = 0.05$ gives $n = (1.96^2 \times 0.30 \times 0.70)/0.05^2 = 323$
- Using $p = 0.50$ when the proportion is unknown gives the largest, safest sample
- For comparing two proportions or means, size depends on the effect you wish to detect, the significance level and the power
- Power is the probability of detecting a real effect; 80% is conventional and 90% is preferable
- An underpowered study is not merely inconclusive — it is arguably unethical, since participants were exposed to burden for a result that could not have been informative
- Apply a finite population correction when the sample is a large fraction of a small population

ADJUSTING THE CALCULATED SAMPLE

Adjustment	When it applies	Formula
Non-response	Always, in surveys	$n_{\text{final}} = n / (1 - \text{expected non-response})$
Design effect	Cluster sampling	$n_{\text{final}} = n \times \text{DEFF (often 1.5-2.0)}$
Finite population	Sample exceeds ~5% of population	$n_{\text{adj}} = n / (1 + (n-1)/N)$
Attrition	Longitudinal designs	Inflate by expected loss to follow-up

Putting it together

- State the sampling frame explicitly, and say what it excludes
- Report the calculation with every input value, not just the final number
- Say who was excluded and why, and how non-response was handled
- Mixed methods studies size each strand by its own logic — power calculation for the quantitative strand, information power for the qualitative
- A nested qualitative sample is often drawn purposively from the quantitative sample, which strengthens integration

Worked examples

Calculating a sample size, then adjusting it honestly

A prevalence survey of anaemia among pregnant women in one district. Previous work suggests roughly 35% prevalence. You want a 5% margin of error at 95% confidence, sampling by health facility.

1. Base formula: $n = Z^2 p(1-p) / d^2 = (1.96^2 \times 0.35 \times 0.65) / 0.05^2$
2. That is $(3.8416 \times 0.2275) / 0.0025 = 0.8740 / 0.0025 = 350$ (rounding up).
3. Cluster sampling by facility: multiply by a design effect of 1.5 $\rightarrow 350 \times 1.5 = 525$.
4. Expect 10% non-response: $525 / 0.90 = 584$.
5. Final target: 584 women. In R: `ceiling(1.96^2 * 0.35 * 0.65 / 0.05^2 * 1.5 / 0.9)`
6. Report every input — Z, p, d, design effect and non-response — not just the number 584.

THE POINT

Reviewers check the arithmetic. A sample size stated without its inputs reads as though it was chosen first and justified afterwards.

Two studies in one document is not mixed methods

A thesis reports a 600-household survey in Chapter 4 and 20 interviews in Chapter 5. The conclusion summarises both. The student calls it a mixed methods study.

1. Ask the integration question: what did combining them reveal that neither gave alone?
2. If the answer is "nothing — they are both about the same topic", it is two studies bound together.
3. Real integration would be: the survey found low clinic attendance in one sub-district; the interviews were sampled from there specifically to explain why.
4. Or a joint display: a table with survey findings in one column and the interview themes that explain or contradict them alongside.
5. The interview sample can be drawn from the survey respondents, which ties the strands together by design rather than by assertion.

THE POINT

Integration is the defining feature. Plan the point where the strands meet at proposal stage — it cannot be added convincingly at write-up.

Key terms

Sampling frame

The list or procedure from which the sample is actually drawn.

Design effect (DEFF)

The factor by which variance increases because of cluster sampling rather than simple random sampling.

Power

The probability that a study will detect an effect of a specified size if it truly exists.

Effect size

The magnitude of a difference or association, independent of sample size.

Integration

The deliberate bringing together of quantitative and qualitative strands to produce insight neither gives alone.

Joint display

A table or figure presenting quantitative and qualitative results side by side to show how they relate.

Activity

Calculate the sample size for a prevalence study in your own field: choose p , set $d = 0.05$, then inflate for 15% non-response and a design effect of 1.5. Show every step. Then state, in one sentence each, the sampling frame and what it excludes.

Further reading

- **Creswell & Plano Clark — Designing and Conducting Mixed Methods Research** — The standard text; strong on integration and joint displays.
- **Fetters, Curry & Creswell (2013) — "Achieving Integration in Mixed Methods Designs"** — The clearest short account of what integration actually requires.
- **Lwanga & Lemeshow — Sample Size Determination in Health Studies (WHO)** — Free from WHO; tables for most common situations.

Self-check questions

1. A team surveys 600 households, finds unexpectedly low clinic attendance, then interviews 20 non-attenders to understand why. This design is:
 - a. Convergent parallel
 - b. Explanatory sequential
 - c. Exploratory sequential
 - d. Embedded
2. What most clearly distinguishes genuine mixed methods from simply doing two studies?
 - a. Using a larger sample
 - b. Deliberate integration of the strands to produce insight neither gives alone
 - c. Publishing both in the same journal
 - d. Collecting both types of data at the same time
3. For a prevalence study with expected proportion 0.30 and margin of error 0.05 at 95% confidence, the required sample is approximately:
 - a. 196
 - b. 323
 - c. 384
 - d. 512

4. Which strategy guarantees that a small subgroup is adequately represented?
- Simple random sampling
 - Stratified sampling
 - Convenience sampling
 - Systematic sampling
5. Cluster sampling requires a larger sample than simple random sampling because:
- Clusters are harder to find
 - Responses within a cluster are correlated, which increases variance
 - It is a non-probability method
 - Non-response is always higher
6. The strongest argument that a seriously underpowered study is unethical is that:
- It will not be published
 - Participants bore burden and risk for a study incapable of producing an informative answer
 - It wastes the researcher's time
 - It cannot obtain ethical approval
-

Answers

- b.** The quantitative strand comes first and the qualitative strand is used to explain its findings. That ordering and purpose define an explanatory sequential design.
 - b.** Integration is the defining criterion. Reporting a survey and some interviews in adjacent chapters, with no point where they speak to each other, is not a mixed methods design.
 - b.** $n = Z^2 p(1-p)/d^2 = (3.8416 \times 0.30 \times 0.70)/0.0025 = 0.8067/0.0025 \approx 323$. The familiar 384 comes from using $p = 0.50$, the most conservative value.
 - b.** Stratification divides the population first and samples within each stratum, so small subgroups are represented by design. Simple random sampling may by chance include very few of them.
 - b.** People within the same village, school or clinic tend to resemble one another, so each observation carries less independent information. The design effect quantifies this and inflates the required sample.
 - b.** Ethical justification for research rests on the prospect of useful knowledge. If the design could not have detected a meaningful effect, participants were exposed to burden with no realistic prospect of benefit.
-